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# Durability and Reliability of Electric Vehicle Batteries under Electric Utility Grid Operations:

# Bidirectional Charging Impact Analysis

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## Abstract

Vehicle-to-grid and Grid-to-vehicle strategies are often cited as promising to mitigate the intermittency of renewable energy on electric power grids. However, their impact on the vehicle battery degradation has not been investigated in detail. The aim of this work is to understand the impact of bidirectional charging on commercial Li-ion cells used in electric vehicles today. Results show that additional cycling to discharge vehicle batteries to the power grid, even at constant power, is detrimental to cell performance. This additional use of the battery packs could shorten the lifetime for vehicle use to less than five years. By contrast, the impact of delaying the charge in order to reduce the impact on the power grid is found to be negligible at room temperature, but could be significant in warmer climates.

Keywords:

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Lithium-ion, Cell-to-cell variations, Vehicle-to-grid, Grid-to-vehicle, V2G, G2V, Smart Grid, Electric vehicle, Bidirectional charging

# 1 INTRODUCTION

Given electric vehicles (EVs) projected penetration into the market [1], their batteries have the potential to provide significant energy storage for the electric power grid in the future by allowing the grid to give and take energy from the batteries when needed [2]. This bidirectional charging is known as Grid-to-vehicle (G2V) and Vehicle-to-grid (V2G) respectively. V2G and G2V strategies are thought to be useful for a range of energy storage applications to supplement the power used by a building; provide a range of services to the power grid including frequency control, demand response, spinning reserves and energy shifting, known collectively as ancillary services; smooth the variable output from renewable electricity generation, and utilize surplus energy when supply exceeds demand [15]. The potential of using EV battery capacity for grid storage has been the subject of much discussion in recent years [3-13] but few studies have tested grid impact on EV battery degradation. To the best of our knowledge, only one battery testing study, Peterson and Whitacre in 2010 [14], was published in open literature. Most battery-related work published so far focuses on modeling [8, 15-21]. Although some studies stand out (e.g. [21]), the models unfortunately do not represent realistic battery degradation.

Understanding the real impact of bidirectional charging on batteries is essential to weigh their viability as a grid support service. Indeed, V2G will induce more usage of the cells by discharging to the power grid and having to recharge the battery again for transportation.

Controlled G2V on the other hand will allow the cells to rest at a different state of charge (SOC) compared with charging immediately upon arrival, and therefore might induce a

different calendar aging. Since battery degradation is path dependent [22], any changes in usage can lead to different degradation mechanisms and drastically influence the reliability of the batteries. Consumers will likely participate only in V2G/G2V programs if they are beneficial, i.e. if the income from "selling" energy at least provides some compensation for any reduction in battery lifetime and performance.

Under the US Department of Transportation funded Electric Vehicle Transportation Center, the Hawaii Natural Energy Institute has partnered with the Florida Solar Energy Center to perform laboratory testing on commercial Li-ion cells to investigate the impact of bidirectional charging on EV type batteries.

Unlike Peterson and Whitacre [14] who used iron phosphate cells, which are not used in any of the main EVs on the market today outside of China [23], we elected to use Li-ion cells with a Nickel Aluminum Cobalt (NCA) chemistry, as used in some EVs available today. In this study, V2G was considered to provide energy shifting to the power grid and other applications will be tested at another time. Additionally, G2V was investigated as a means to shift the power demand (load), and limit the use of costly operating reserves and peaking plants.

In order to assess the impact of the V2G and G2V on the cells it is important to take into consideration all the different ways the cells can be used. Because battery degradation is path dependent, it is possible that in some cases more intense cell use may still result in lower degradation [24, 25]. As an example, performing V2G during times of peak demand for electricity will also delay the battery charge. It is therefore important to separate and test the impact of the delay by itself. To take these issues into account, the effects of V2G/G2V combined with different charging schedules (1 or 2 charges a day,

immediate or delayed charging) and different charging currents (level 2 or fast charging) were investigated. Moreover, the effect of calendar aging at different temperatures was also investigated in a second set of experiments.

A well-controlled experimental design was essential to separate the contribution of each factor, as well as to eventually determine combined effects. There are multiple methodologies to design controlled experiments [26, 27] and some were previously and successfully applied to battery testing [28-30]. A screening design was used in this study to investigate the impact of V2G and G2V because of the categorical nature of the variables, i.e. different duty cycles and not different intensities. For the complementary calendar aging experiment, a surface response design was selected to allow interpolation between selected data points and to gain a better understanding over the entire range of temperature and SOC.

## 2 EXPERIMENTAL

A batch of 100 Panasonic graphite (GIC) / $LiNi_xCo_{1-x-y}Al_yO_2$ (NCA) NCR18650B cylindrical battery cells rated at 3350 mAh were used in this research since similar to those used in some EVs [23]. Cell details and conditioning and characterization tests as well as the cell-to-cell variation analysis [31] were presented in part 1 of this work [32]. Using results from the cell-to-cell variation distribution [32], the 36 cells chosen for the cycling study were within the outlier boundaries of the variation distribution. For the calendar aging experiment, 16 cells were selected on the basis of their very high consistency (located within the quartile boundaries for each of the three attributes of cell-to-cell variation) [32]. Since a smaller number of samples were tested for the calendar aging study, a tighter selection criteria was used.

As described in our previous work, all cells were subjected to the initial conditioning and characterization test prior to the beginning of the cycling experiment [32]. Following the characterization test, and before the start of the duty cycle experiment, a reference performance test (RPT) was conducted to determine the cell performance. It consisted of charge and discharge cycles of 35 hours (C/35), 5 hours (C/5), and 3 hours (C/3). The RPT procedure, described in detail in part 1 of this work [32], included some remnant capacity measurements at the end of each regime as well a 4 hour rest prior to and after the remnant capacity measurements. The rate capability was calculated at each RPT by dividing the C/3 capacity by the C/35 capacity. The resistance was calculated from the slope of the ohmic drop associated with the transition from a rested stage to a C/3 discharge rate using Ohms' law. The testing was typically interrupted to perform the RPT every 6 weeks for the cycling experiment and 8 weeks for the calendar aging experiment. All RPTs and cycling experiments were performed at 25°C and the calendar aging was carried out at -27°C, 25°C, 45°C and 55°C.

Equipment used for all tests was a calibrated 40 channel ARBIN LBT-5V-25A machine and the following environmental chambers:

- Norlake Scientific LF201WWW/0 freezer for the -27°C experiments,
- Amerex IC500-R chambers for the 25°C (calendar, RPTs for calendar aging experiments and cycling) and 45°C experiments.
- An Espec BTZ475 chamber for the 55°C experiments.

## 3 DESIGN OF EXPERIMENT

Since battery degradation is extremely sensitive to usage, it is necessary to test the cells in conditions as close as possible to an actual duty cycle. In order to replicate the real usage of an EV, a day of usage was decomposed into several steps: The EV started at home (@Home), was driven to work, parked at work (@Work) and was later used to commute back home. Based on this daily schedule, the following hypotheses were used to build the different duty cycles for this study:

*Driving Cycle Protocol*. The driving cycle protocol (to and from work) was based on scaled down current from real driving data selected from an HNEI in-house database [33-35]. A one hour round trip, Figure S1, was chosen with every type of road condition from stop-&-go driving to highway driving [32-34]. The database contains 20 other variations of this commute and this could offer opportunities for follow-up studies.

*V2G protocol*. As mentioned in the introduction, V2G for ancillary service was investigated. Because of the relative size of the EV battery compared with the power grid, a constant power profile was assumed for the duty cycle. This constant power rate will be assumed to be the maximum deliverable by a 25kWh battery pack and the infrastructure available in most homes (240V, 30A) [14]. Based on these assumptions, the home charger can maintain at most 7.2 kW for energy transfer, and thus recharges at most one third of the battery capacity every hour. Accounting for the typical slight overbuilding of commercial battery packs, the constant power rate chosen for the V2G protocol was slightly lower than the maximum, and set at one quarter of the rated power (P/4).

The duration of the V2G step was based on a power grid representative of an average North America grid with two price peaks that last approximately one hour each, at noon and in the early evening [4]. Maximum monetary benefits will be achieved by selling as much electricity as possible during these two peak periods and charging the EV when electricity price is the lowest.

According to these assumptions, testing was designed to apply a P/4 discharge step for one hour, up to twice a day depending on the charging protocol.

*Charging Protocol.* Typical home battery chargers can usually charge an EV battery pack in 21 hours with level one charging and in about 8 hours for level two charging [36]. On-board chargers, such as the one in the Nissan Leaf [37], allows faster charges in four hours or less [38]. For this study, we assumed level two charging and an eight hour charge for the @Home charging and access to a fast charger to fully charge the vehicle in four hours @Work. It was also assumed that the battery was charged at least once a day depending on the duty cycles tested.

*Experimental design*. This study was used to investigate the effect of constant power V2G and G2V protocols on single cells. Since V2G and G2V protocols might have a combined effect with other variables such as the charging schedule, and the charging level, it is essential to study a matrix of tests encompassing the different combinations.

Four different duty cycles were selected to occur during the @Home and @Work periods: A V2G discharge followed by a charge and rest (DCR), a charge then a rest (CR, standard usage), a rest then a charge (RC, G2V delayed charging), and just a rest (R).

These four duty cycles can be visualized in a $4^2$ multi-categorical fractional design of experiment [26, 27] with 2 factors and 4 combinations. The two factors are the parked periods (@Home and @Work) and the four combinations are the duty cycles (DCR, CR, RC and R). This design of experiment is summarized in Figure 1(a) and consists of 16 experiments. Experiments were noted with the @Work duty cycle and the @Home duty cycle, such as CR-DCR. The number of experiments was reduced to 15 since at least one charge per day was assumed, thus eliminating the R-R experiment. These 15 experiments with four different combinations of duty cycles allow determination of the effects of V2G (DCR vs. CR and RC), of G2V (CR vs. RC), and of two charges per day (R-* vs. CR-* and RC-*). Finally, the effect of the charging power was also considered since the @Home and @Work charging schemes are different (*-R vs. R-*).

The next issue to address is reproducibility. In order to have confidence in the observed effect, and since the variability in aging of these cells is unknown [39, 40], experiments were repeated 3 times, increasing the total from 15 to 45. It was further assumed that the car was charged @Home every day. As a result, three duty cycles (DCR-R, CR-R and RC-R) were removed from the test matrix and therefore the impact of the charging level (regular vs. fast) was no longer investigated. This reduced the number of experiments to 12 and the number of cells to test to 36.

*Test acceleration.* EV batteries are expected to last more than 8 years and at least 100,000 miles [41]; hence, any cycle aging experiments must be accelerated to be completed in a reasonable amount of time. In order to accelerate the aging, the duration of the duty cycle was shortened so that it could be repeated several times a day for the experiment. This

was first done by eliminating the eight hours of common rest time (i.e. the time where the cells were at rest in all duty cycles at the same time). Second, the charge duration was optimized. The full charge time for each duty cycle would theoretically total 12 hours (4 hours @Work + 8hours @Home) but, since the cells were not completely discharged by the commute distance, the actual charge duration would be much shorter. Each driving leg consumed about 15% of the cell capacity. Additionally, the V2G steps shall expend 25% to 30% of the cell capacity (1 hour at a P/4). The cells were therefore never discharged more than 45%; 15% from driving and 30% from V2G. The time taken to fully charge was consequently less than half of the time made available for charging. Therefore, reducing the time devoted to the charge by a third will allow the test to be significantly sped up while leaving some room accounting for an eventual increase of the constant voltage step duration after aging and fading power response. To ensure the cells were at similar states prior to each driving leg regardless of the duty cycle performed, a 25-minute rest was added to the schedule after a charge. A 5-minute rest prior to a V2G or charging step was also included to mimic the time it takes to plug the car in and enter information into the charging station.

With all of these adaptations, the testing time for each loop was reduced from 24 hours to 11 hours, with one hour of driving and 10 hours of plugged in time, Figure 1(b,c). This allowed testing the impact of 65 “days” in 30 days, more than twice as fast as real life. In order to make sure that the test will stay in sync, all steps were time limited, Figure 1(b). Figure 1(c) presents the detail of the 12 duty cycles of the cycling experiment on pristine cells.

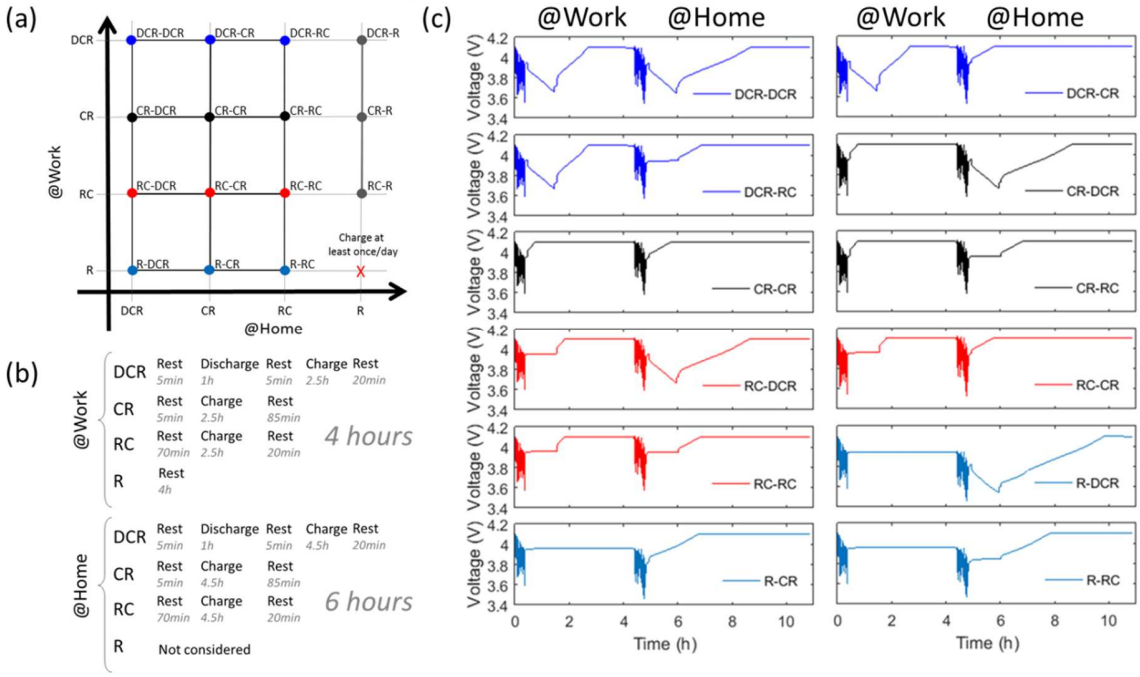



Figure 1: (a) Test matrix, (b) @Work and @Home duration detail, and (c) detail on the 12 duty schedules used in this study on pristine cells.

*Calendar aging.* With the cycling experiment daily cycles reduced from 24 hours to 11 hours, it becomes essential to also investigate calendar aging to account for the impact of the 13 hours of rest time that were eliminated from the cycling experiment. The impact of different temperatures on calendar aging was also assessed. The pace of calendar aging is known to be dependent on the cells' SOC and on the environmental temperature [42]. For EV applications, the SOC could be anywhere from 100% to 0% and the temperature, dependent on the location and climate (from -27°C to 55°C in this study).

To test the combined effects across these temperature and SOC ranges would be a prohibitive number of tests, but the number of experiments can be reduced by using the design of experiment response surface methodology to optimize the SOC and

temperature combinations. According to a classic response surface design strategy, the 'central composite' design of experiments [27], only 9 tests are needed to cover the entire range, Figure 2(a). Since calendar aging on GIC//NCA cells occurs for the most part at temperatures above room temperature and at a higher SOC [42], the density of experiments in a central composite design is not optimal and it was decided to increase the density of experiments in the top-right quadrant in order to study the temperature and SOC range of interest, and to increase the accuracy of the final model. Temperature was defined as a discrete factor with 4 levels based on the available environmental chambers (-27°C, 25°C, 45°C and 55°C) to design a custom response surface plan with computer assisted methods using the Design Expert® 9 software [43]. The selected design is presented in Figure 2(b) and it consists of 8 experiments focusing on high temperature and high SOC and, for reproducibility purposes, two cells will be tested with each condition. It has to be noted that, because of a machine issue, the last rate of the last RPT could not be recorded for 3 cells. These values were therefore not reproduced.

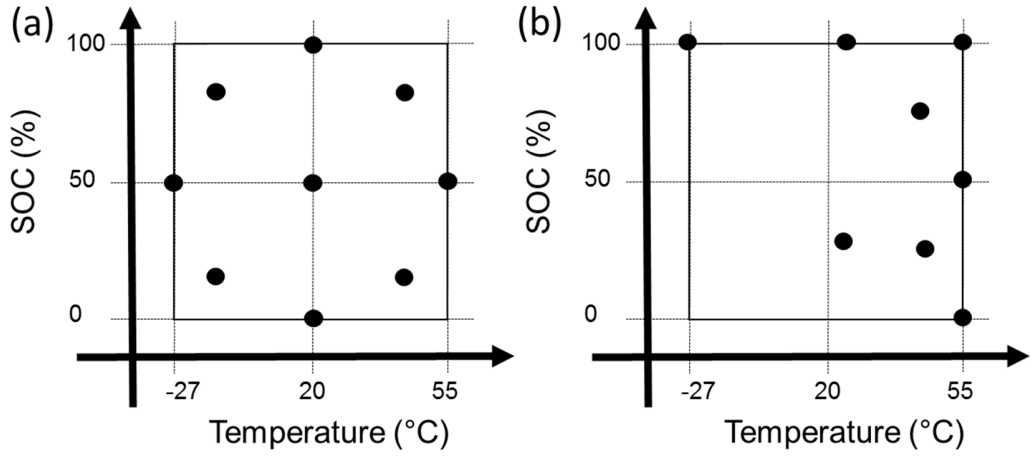


Figure 2: Calendar aging test matrix for (a) a central composite design and (b) the optimized design.

# 4 RESULTS

## 4.1 Impact of cycle aging

The impact of cycle aging was investigated using the same methodology as that of the cell-to-cell variation analysis [31]. The changes in maximum capacity, resistance and rate capability were characterized in order to assess performance variations induced by the different cycling duty cycles.

Figure 3 presents the results of the cycling experiment with (a) the evolution of the capacity loss and (b) the resistance as a function of the equivalent months of driving, while (c) presents the evolution of the capacity loss as a function of the capacity throughput.

After 18 equivalent months of driving, the cells lost from 5% to 9% of their initial capacity, Figure 3(a). The cells that experienced the highest loss were the ones that underwent V2G twice a day; while the cells that lost the least were the cells that were subjected to no V2G and that were charging twice a day. The results of the cycling experiment are consistent as shown by the length of the error bars, corresponding to the capacity spread between the three cells undergoing the same duty cycle, which is below 1% of the cell normalized capacity on Figure 3. Throughout the experiment, all rate capabilities remained within 1% of the initial values (not shown) and the ohmic resistance of the cells increased by 40% on average, Figure 3(b). The cells that

experienced V2G had their resistance increased slightly more (45% on average) than the others (35% on average), but the distribution was narrow at only 10% after 18 months of equivalent cycling.

From Figure 3(a), it appears that the cells that were performing V2G twice a day lost capacity faster than the others, but they were also used more heavily, with 75% of the cell capacity in total consumed every day. The cells that were subjected to V2G once a day used around 50% of their capacity daily and the cells that only experienced driving and charging only used 30%. To take this into account, cell degradation was plotted as a function of the exchanged capacity, Figure 3(c). This plot revealed a tighter distribution of the capacity loss values. Indeed, after a total of 500Ah exchanged (150 equivalent full cycles), the capacity loss of all 36 cells was within 1%. After 900Ah exchanged (270 equivalent full cycles), the 18 cells that reached that point were also within 1%.

It is interesting to note that the cells that degraded the fastest in terms of equivalent full cycles are not the same as those that lost the most capacity in terms of equivalent months of usage. In terms of exchanged capacity, the cells that degraded the fastest were the ones that experienced a rest for the @Work period and no V2G for the @Home period. This is likely to be an effect of calendar aging since those cells spent up to 80% more of the testing time at rest relative to the other cells.

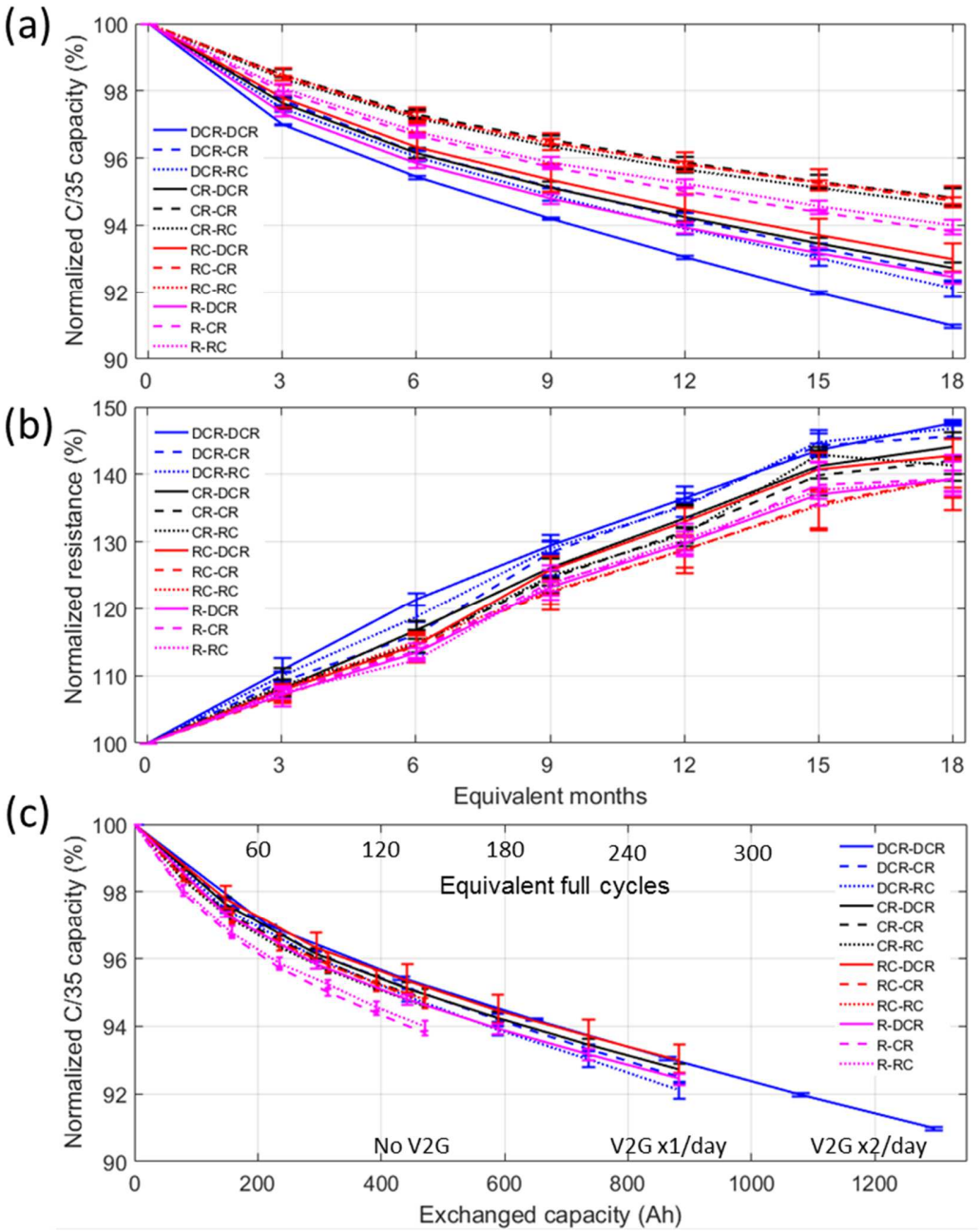



Figure 3: (a) Average capacity fade and (b) resistance changes vs. equivalent months of cycling; (c) average capacity fade vs. capacity throughput for the 12 duty cycles tested in this work. Error bars indicate the spread of values between the three tested cells per condition.

The observed performance degradation in this study was moderate compared to what has been published in literature for GIC//NCA cells. Frisco at *al.* [44] used the exact same commercial battery under C/3 constant current cycling conditions reported 15% capacity loss and a 100% resistance increase after 350 cycles compared to at most 10% and 50% respectively in this study. According to studies on other commercial NCA cells with different specifications, the capacity loss reported for 350 cycles for constant current cycling varied from 1% to 15% [28, 45, 46] and went up to 20% for a high rate pulse duty cycle [45].

### 4.2 Impact of calendar aging

The same analysis procedure as that used for duty cycling was employed for the calendar aging experiment. Figure 4 presents the evolution of (a) capacity loss, (b) rate capability and (c) ohmic resistance as a function of time for the different temperature/SOC combinations.

*Capacity.* The cells lost between 1% (-27°C/99% SOC) and 13% (55°C/81.5% SOC) of their initial capacity after 66 weeks of storage. Therefore temperature clearly has an impact on cell degradation. There is also a significant impact from the SOC as, at equal temperature, the cells at a lower SOC lost less capacity. At 55°C, increasing the SOC from 5% to 81.5% induced 6.5% more capacity loss and 4% was induced between 20% and 70% SOC at 45°C. At room temperature, the 66 weeks calendar aging induced close to 3% loss at 50% SOC and slightly above 4% loss at 100% SOC. Similar to the cycle aging experiment, the results are consistent with a maximum of 0.35% difference in

degradation between cells undergoing the same aging conditions. Finally, it has to be noted that results are in the same order of magnitude as those reported in the literature for other types of NCA-based commercial cells [47].

*Rate capability.* The evolution of the rate capability for the calendar aging experiment, Figure 4(b), appeared complex. The rate capability decreased with time for the low temperature experiments, was stable at room temperature (at 50% SOC) and increased for the high temperature experiments. Moreover, increasing the storage SOC has a negative effect at low temperature (further reducing the rate capability) but a positive effect at high temperature (further increasing the rate capability). All the RPTs were performed at room temperature, and the observed variations are therefore the result of the temperature excursions and not an effect of the actual cell temperature during the RPT.

*Resistance.* The evolution of the resistance with storage time is presented in Figure 4(c). The cells stored at 55°C/81.5% SOC exhibited the largest increase in resistance, 110%; while, the cells that were stored at room temperature and 50% SOC exhibited the lowest, 20%. These increases suggests that both low and high temperature are factors in the resistance increase. Further, a higher SOC also increased the resistance. The results were consistent except for the -27°C/99% SOC cells where one cell had a resistance increase comparable to the 25°C/100% SOC cells and one to the 55°C/81.5% SOC cells. Overall, the resistance evolution was similar to the trends reported in the literature on other types of NCA-base commercial cells [48, 49].

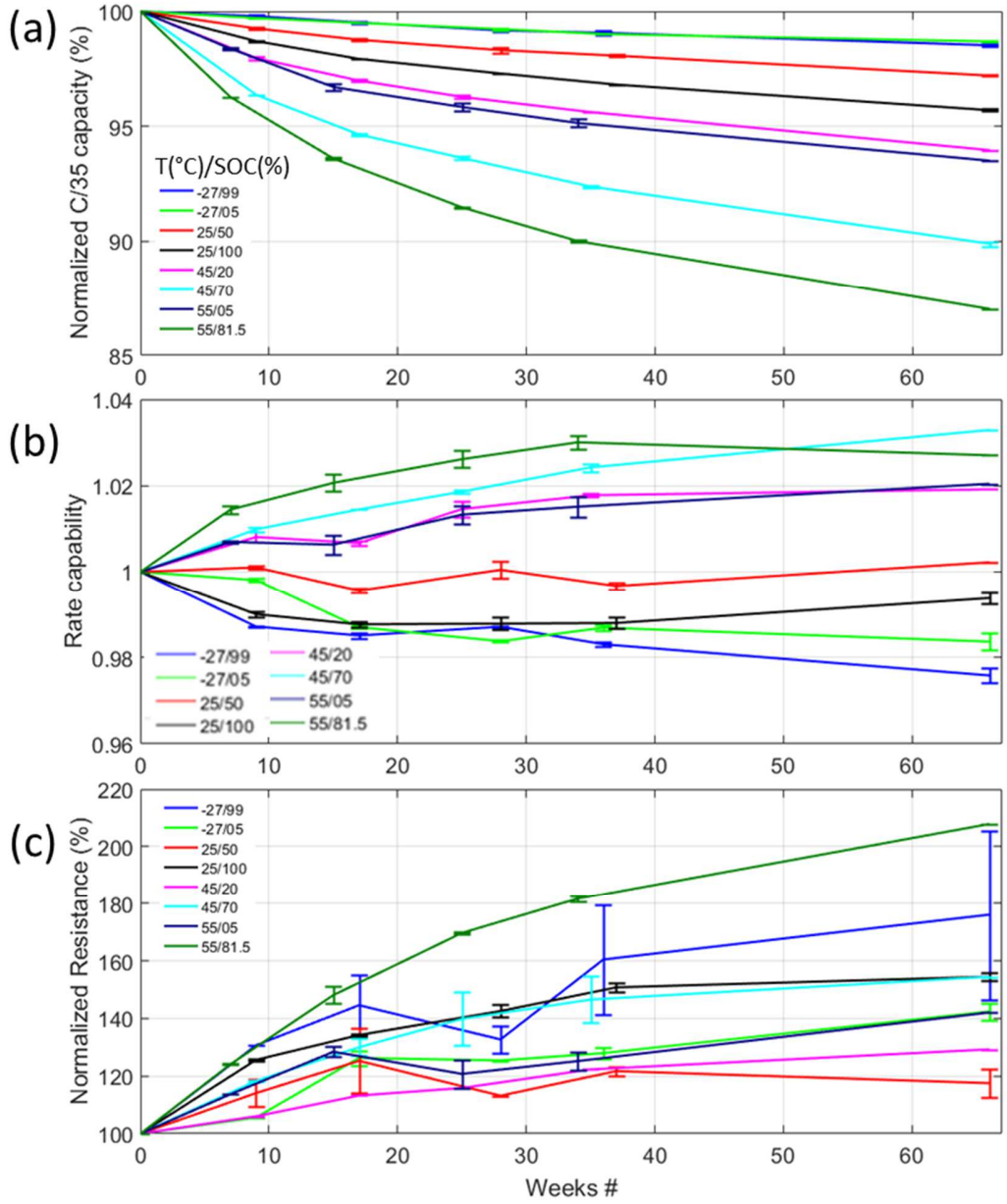



Figure 4: (a) Average capacity fade, (b) rate capability variations, and (c) resistance variations vs. temperature and SOC for the eight calendar aging conditions tested in this work. Error bars indicate the spread of values between the two tested cells per condition.

Finally, self-discharge during the calendar aging periods was investigated via the evolution of the rest cell voltages from the beginning to the end of each calendar aging period (not shown). The small potential variations detected after 8 weeks and after 30 weeks were similar. In addition, the potential was found to decrease if the cell reached its open circuit voltage (OCV) from a charge step. The potential increased if it reached its OCV from a discharge step. These findings imply that the potential variations were induced by the relaxation process. No significant self-discharge was observed.

# 5 DISCUSSION

## 5.1 Cycling experiment effect analysis

The different factors (i.e. testing combinations) associated with capacity loss are presented in Figure 5(a) as calculated from an analysis of variance (ANOVA), a statistical method used to analyze differences between means. Calculations were performed using Design Expert® software [43] and coded variables have been plotted. Coded variables allow comparison of the effects of different factors on a common scale to determine which factor has the largest impact on the response. It also ensures the orthogonality of the model which eases the analysis. From the large F-values and low P-values (< 0.1%), all the factors and the resulting model were deemed significant for all aging conditions. Figure 5 presents a graphical analysis of the ANOVA parameters. Figure 5(a) shows that the two parked periods, A (@work) and B (@ home), were more significant than their interactions, AB. As seen in the figure, the first factor (1 = DCR)

had the greatest impact on capacity loss as it induced up to 1.5% additional capacity loss compared to the mean degradation. Factors 2 and 3 (CR and RC) produced the opposite effect decreasing capacity loss compared to the mean. Factor 4 (R) caused relatively little change to the average capacity loss. Figure 5 (a) also shows that all the combined effects (in red) were, for the most part, small, but increased with the number of cycles. The term AB was found to be significant only after 18 equivalent months of aging. Before that, although the F-values and P-values were gradually increasing and decreasing, respectively, they were not significant compared to the noise. The only slightly significant interaction, A[1]B[3], suggested that the degradation induced by the third factor @Home (RC) was slightly higher when the first factor was performed during the @Work period (DCR).

The ANOVA results on uncoded variables verified observations from the cycling experiment (presented in Figure 3), and allowed us to determine that all the capacity loss followed a similar power law, $y=ax^{0.6}$, versus time, and that they were therefore approximately linear on a month$^{0.6}$ scale. The results were compiled in a bar plot for better visualization, Figure 5(b), and the fitting parameters are reported in Table 1. In this representation, it is clear that the DCR sub-cycles (first column and last line) whether performed @Work or @Home, caused faster capacity fade than the other test factors. The duty cycles with no V2G steps and two charges per day induced the slowest capacity fade.

Table 1: Fitting parameters for the $y = ax^{0.6}$ model with standard deviation in parenthesis.

| | $R^2$ | $a$ |
|---|---|---|
| DCR-DCR | 0.999 | 1.573 (±0.008) |

| | | |
|---|---|---|
| DCR-CR | 0.996 | 1.308 (±0.016) |
| DCR-RC | 0.998 | 1.374 (±0.011) |
| CR-DCR | 0.998 | 1.290 (±0.010) |
| CR-CR | 0.992 | 0.917 (±0.015) |
| CR-RC | 0.994 | 0.960 (±0.014) |
| RC-DCR | 0.998 | 1.238 (±0.011) |
| RC-CR | 0.995 | 0.932 (±0.012) |
| RC-RC | 0.990 | 0.928 (±0.017) |
| R-DCR | 0.995 | 1.358 (±0.016) |
| R-CR | 0.996 | 1.110 (±0.012) |
| R-RC | 0.995 | 1.071 (±0.013) |

In order to quantify the impact of the different combinations, we can use the data shown in Figure 5(b) to isolate and further quantify the impact of the variables: V2G protocol (twice a day vs. once a day vs. none), the impact of one charge per day vs. two charges per day, and the impact of delayed G2V charging over immediate charging. Performing the V2G step twice a day accelerated the capacity loss by 75%, while V2G only once a day accelerated the loss by 33%. Charging twice a day instead of once slowed the capacity loss by 5%. Replacing an immediate charge with a delayed charge (the G2V step) had no significant effect (less than 1% difference in capacity loss).

The previous paragraph was devoted to the analysis of the capacity loss on a time scale. Looking at the same degradation on an exchanged capacity scale, Figure 5 (c) presents the ANOVA results after 500 Ah exchanged. The cells that lost the most capacity were charged once per day (purple bars). Cells that degraded the least, (25%) experienced a V2G step twice per day. This suggests that increasing the use of the cells slowed the rate of degradation on the exchanged capacity scale. This effect was likely due to the

increased impact of calendar aging where degradation occurs without any usage of the cells.

The same ANOVA analysis was performed on the resistance data on the time scale. Figure 5(d) presents a bar plot summarizing the resistance increases after 18 months of equivalent driving. Just as for the capacity loss, the V2G schedule was responsible for the largest resistance increase. However, as opposed to the capacity loss analysis, the one charge per day cycle affected the metric the least. There was also an observable difference between the results from delayed G2V and immediate charging protocols. The results show that a higher resistance increase occurred when the cell was charged right away. It should be noted that no correlation was found between the capacity loss and the resistance increase (Spearman correlation coefficients $< 0.45$ at any equivalent month).

To summarize, for this cell characteristics and under a 1-hour P/4 constant power protocol, the V2G step had a large impact on capacity loss as well as resistance increase, making vehicle-to-grid strategies disadvantageous to EV owners. Therefore, equitable incentives would need to be given to utility customers a compensation for the reduction in battery life. On the other hand, delayed G2V had no impact on the capacity loss and produced less of a resistance increase. Hence, application of delayed charging could be mutually beneficial and adopted extensively if utility customers are properly educated. Surprisingly, topping off the car during the day (i.e. charging @Work and @Home) was beneficial for the batteries in terms of capacity loss, but not for the resistance. This result could be an effect of calendar aging, but needs more rigorous electrochemical analysis, possibly as part of a future study, to ascertain its origin and overall impact on the battery.

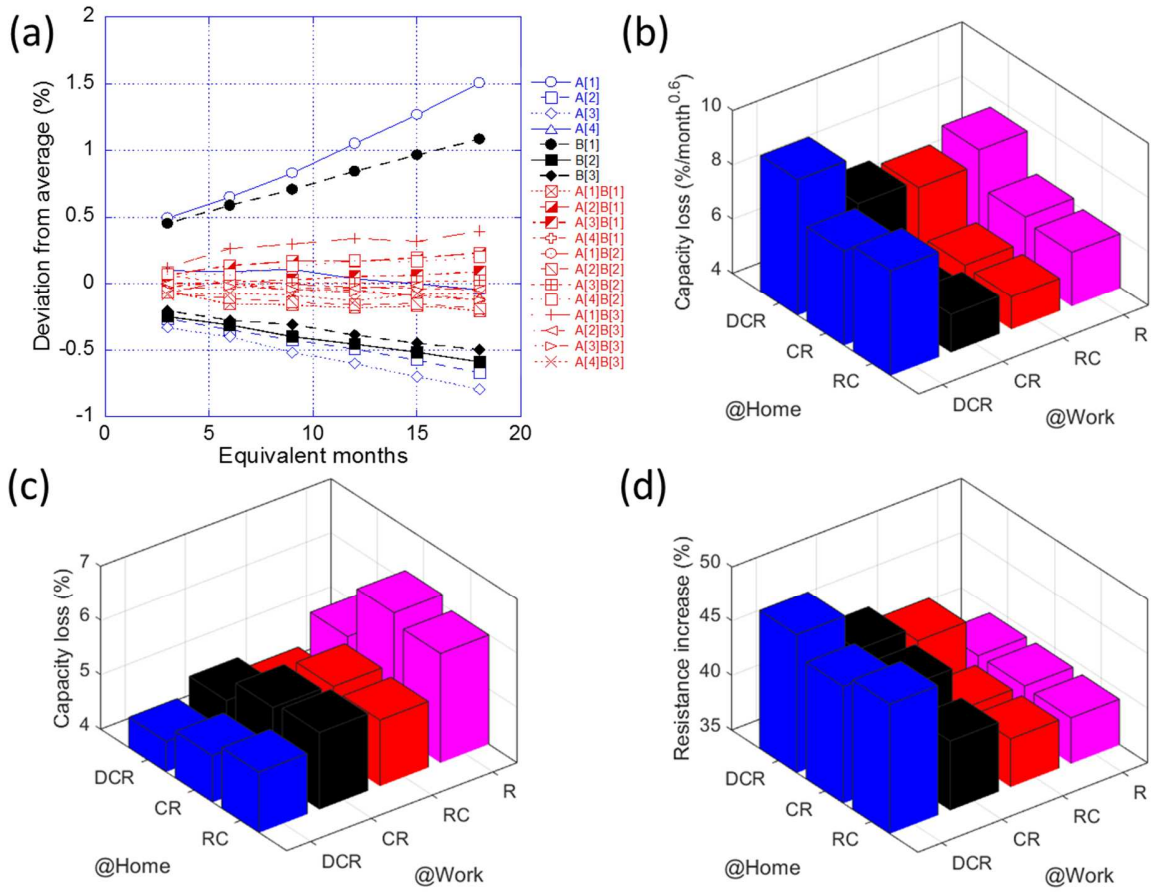



Figure 5: (a) Evolution of coded factors with aging, (b) bar plot showing the pace of capacity loss, (c) comparison of the impact on capacity loss of V2G twice a day, V2G once a day, no V2G, regular charging, delayed charging (G2V), 2 charges per day and 1 charge per day, and (d) bar plot of the resistance increase after 18 months of aging.

## 5.2 Calendar aging experiment effect analysis

The capacity loss ($Q_{Loss}$, percentage) after 66 weeks of aging as a function of temperature (T, Celsius degrees) and SOC (percentage) is presented in a 3D plot in Figure 6(a). Out of several models evaluated, a double-quadratic model (equation 1) was the best fit with an average goodness of fit ($R^2$) of 0.95 and P-values below 0.01% for all the data points at any time (t, weeks), Figure 6(b).

$$Q_{loss}(t) = p_0 + p_1 T + p_2 SOC + p_3 TSOC + p_4 T^2 + p_5 SOC^2 \quad (1)$$

The ANOVA calculations also confirmed the model, and all the terms were significant throughout the aging process. This model offered the opportunity to compare the weight of the coded parameters ($p_n$) from equation 1 and therefore to analyze their relative impact, Table 2 and Figure 6(c). By comparing ($p_1T + p_4T^2$) to ($p_2SOC + p_5SOC^2$) it can be seen that the high temperature (open end of the parabola) had a much greater impact than high SOCs on the capacity loss (vertex of the parabola) and that this impact is increasing with time. Moreover, the combined effect of temperature and SOC ($p_3$) was significant throughout this study and increasing with time.

Table 2: Double-quadratic model coded parameters with standard deviation in parenthesis.

| | *9 weeks* | *18 weeks* | *27 weeks* | *36 weeks* | *62 weeks* |
|---|---|---|---|---|---|
| $R^2$ | 0.924 | 0.975 | 0.992 | 0.990 | 0.988 |
| $p_0$ | 0.91 (±0.29) | 1.06 (±0.27) | 1.24 (±0.19) | 1.46 (±0.26) | 2.05 (±0.39) |
| $p_1$ | 1.46 (±0.17) | 2.55 (±0.15) | 3.22 (±0.11) | 3.79 (±0.14) | 5.00 (±0.25) |
| $p_2$ | 0.71 (±0.16) | 1.10 (±0.15) | 1.48 (±0.11) | 1.73 (±0.14) | 2.33 (±0.25) |
| $p_3$ | 0.57 (±0.17) | 0.86 (±0.15) | 1.15 (±0.1) | 1.39 (±0.14) | 1.76 (±0.24) |
| $p_4$ | 1.27 (±0.28) | 2.29 (±0.25) | 3.05 (±0.18) | 3.59 (±0.24) | 4.77 (±0.39) |
| $p_5$ | -0.82 (±0.39) | -0.90 (±0.36) | -1.04 (±0.26) | -1.22 (±0.34) | -1.67 (±0.52) |

The double-quadratic model parameters appeared to vary linearly ($|R^2|>0.98$) with the square root of time [50], Figure 6(d). Equation 1 was then generalized to calculate the capacity loss as a function of time, temperature and SOC, equation 2. The modified double-quadratic model was successful in predicting the capacity loss for the entire dataset with errors on average of 0.25% and at most 1%. The fitting parameters and associated standard deviations are presented in Table 3.

$$Q_{loss} = p_{01} + p_{02}t^{0.5} + (p_{11} + p_{12}t^{0.5})T + (p_{21} + p_{22}t^{0.5})SOC + (p_{31} + p_{32}t^{0.5})TSOC + (p_{41} + p_{42}t^{0.5})T^2 + (p_{51} + p_{52}t^{0.5})SOC^2 \quad (2)$$

Table 3: Equation 2 coded fitting parameters with standard deviation in parenthesis.

| | $R^2$ | $p_{*1}$ | $p_{*2}$ |
|---|---|---|---|
| $p_{0*}$ | 0.993 | 0.18 (±0.03) | 0.33 (±0.12) |
| $p_{1*}$ | 0.964 | 0.77(±0.04) | -0.80(±0.16) |
| $p_{2*}$ | 0.986 | 0.34 (±0.02) | -0.33 (±0.07) |
| $p_{3*}$ | 0.986 | 0.27 (±0.02) | -0.27 (±0.7) |
| $p_{4*}$ | 0.989 | 0.78 (±0.03) | -1.04 (±0.12) |
| $p_{5*}$ | 0.958 | -0.13 (±0.03) | -0.39 (±0.13) |

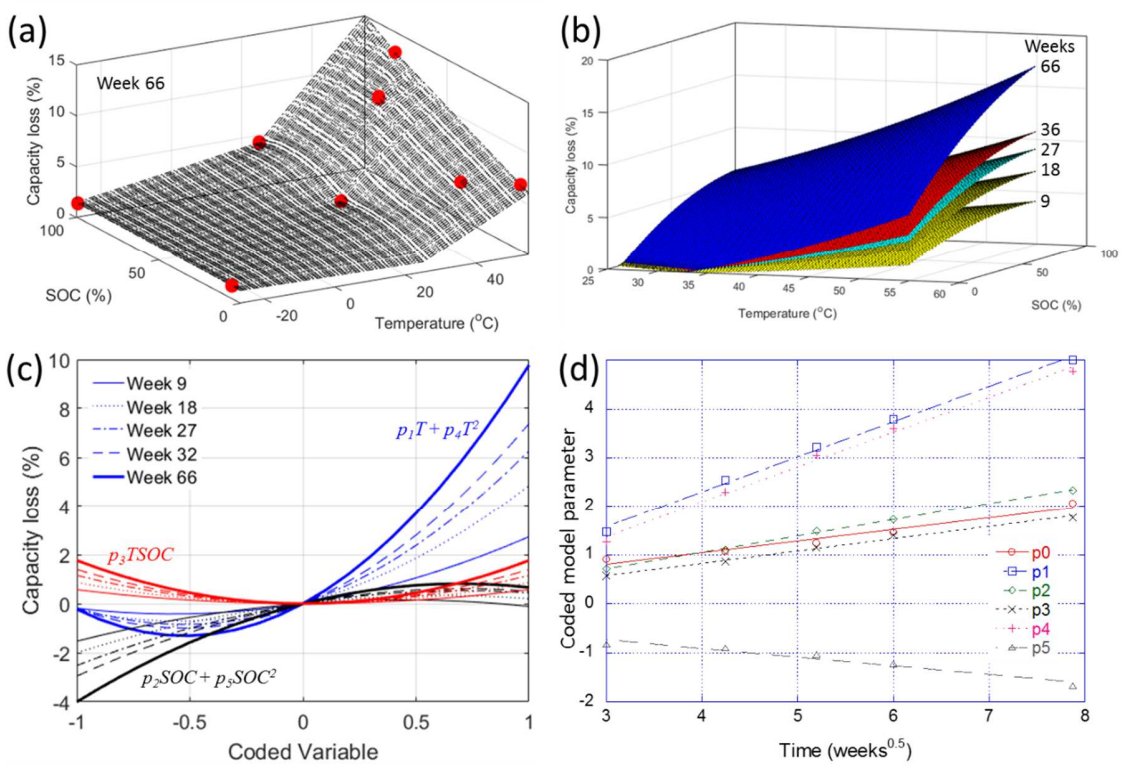



Figure 6: (a) Capacity loss vs. temperature and SOC after 66 weeks of aging. Red points represents the experimental data and the grey mesh a linear interpolation between them. (b) Evolution of the capacity loss as a function of temperature, SOC and time according to a double-quadratic model. (c) Impact of temperature, SOC and their combined effect on capacity loss after 66 weeks. (d) Evolution of the six parameters of the quadratic models with aging.

A suitable ANOVA model could not be determined for resistance and rate capability. Equation 2 was close, but the average error was 8.5% for resistance variations. Nonetheless, insights were gained from the data, Figure 7(a,b). The resistance was clearly more influenced by the storage SOC than the storage temperature, Figure 7(a). At 5% SOC, the resistance increase was independent of the temperature. The data for the

resistance variations formed a bell shape with a minimum near room temperature. At -27°C/99%SOC, there was significant differences between the two tested cells (therefore the data point is presented with an error bar). It may be reasonable to conclude the accurate resistance is the lower of the two, considering its proximity to the experimental value at 25°C and the same SOC. Looking back at Figure 5(d), this is consistent with our observations where the resistance increased more for the immediate charge than for the delayed charge, where the rest was performed at lower SOC.

In contrast, rate capability, a proxy for the cells kinetics, was more influenced by temperature, Figure 7(b). Low temperature degraded the cells kinetics whereas high temperatures improved it. Although it was tempting to explain the degradation of the kinetics and the capacity loss by the increase in resistance, that was not the case here. Indeed, a correlation analysis showed no connection between the resistance increase and the rate capability or the capacity loss, Figure 7(c), with Spearman correlation coefficients below 0.25 and P-values above 0.5. This result was verified by the analysis of the voltage signature of the cell with the highest resistance increase, Figure 7(d). The incremental capacity intensity went back to zero when the discharge cutoff voltage was reached. This implied that the increase in resistance did not push any capacity out of the voltage window at C/35 nor C/3 and therefore that no capacity was lost because of the resistance increase. This conclusion was also true for the cycling experiment.

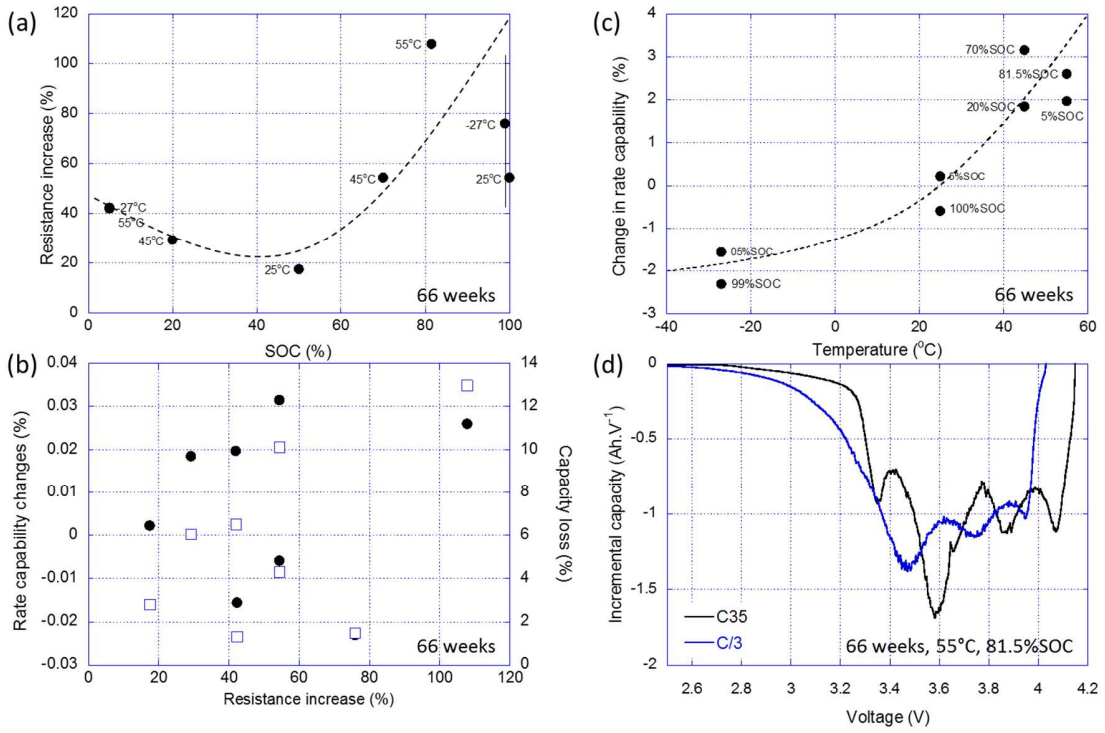



Figure 7: Evolution after 66 weeks of (a) the resistance as a function of SOC and (b) the rate capability as a function of temperature; (c) correlation between capacity loss, rate capability and resistance and (d) incremental capacity curves at C/35 and C/3 for the cell aged at 55°C and 81.5% SOC.

Variations of rate capability and resistance with temperature and SOC most likely originates in differences in the cell degradation mechanism. Although it is well documented that lower temperature slows the cell kinetics, and that higher temperature improves it, this was not applicable here since all RPTs were performed at room temperature. Kinetic improvements were shown also to be a side effect of both loss of lithium inventory and electrochemical milling [51, 52]. In this case, electrochemical milling was unlikely since the cells were all cycled for the same time at the same

temperature, and it should therefore be visible for all cells independent of storage conditions. Loss of lithium inventory is a chemical parasitic reaction [53-55] that is more active at high potential and high temperatures. Hence, it was the most probable culprit to explain capacity loss with both rate capability and resistance increase. Loss of lithium inventory was identified in the literature as the main degradation mechanism for NCA-based cells during high temperature calendar aging [47, 56, 57]. Other causes for resistance increase and rate capability decrease, without significant capacity loss at low temperature, are unclear at this time. Investigating the incremental capacity response of the cell in detail may elucidate the origin of this phenomenon.

### 5.3 Prognosis

From the analysis presented in the previous section, models to forecast the capacity loss due to cycle aging and calendar aging were developed. These prognosis models were used to extrapolate capacity loss for longer periods of time, Figure 8. After 5 years of cycling, the duty cycle that performed a V2G step twice a day is predicted to induce slightly less than 20% capacity loss, whereas the duty cycles with no V2G steps are predicted to induce half of that loss, around 10%. After 5 years of calendar aging, the high temperature and high SOC storage conditions are predicted to result in more than 30% loss of capacity. The cells aged at room temperature are calculated to only experience between 7 and 10% capacity loss.

Since the actual usage of the cells will involve both the cycle aging (11 hour/24 hour) and calendar aging (13 hour/24 hour), the real degradation will be dependent on the combined

effects of both cycling and calendar aging. Given the cycle aging test acceleration scheme, 3 years of calendar aging need to be added to the 5-year cycle aging prediction to be able to account for a 24-hour day. These 3 years of calendar aging alone could induce more than 15% capacity loss at high temperature and SOC, which indicated that these batteries would reach end-of-life well before 5 years. Assuming room temperature aging, the total calendar aging loss to add to the cycle aging will be approximately 6%, depending on the storage SOC. This estimate suggests that implementing a V2G step, even once a day, would cause the battery to lose more than 20% total capacity after 5 years.

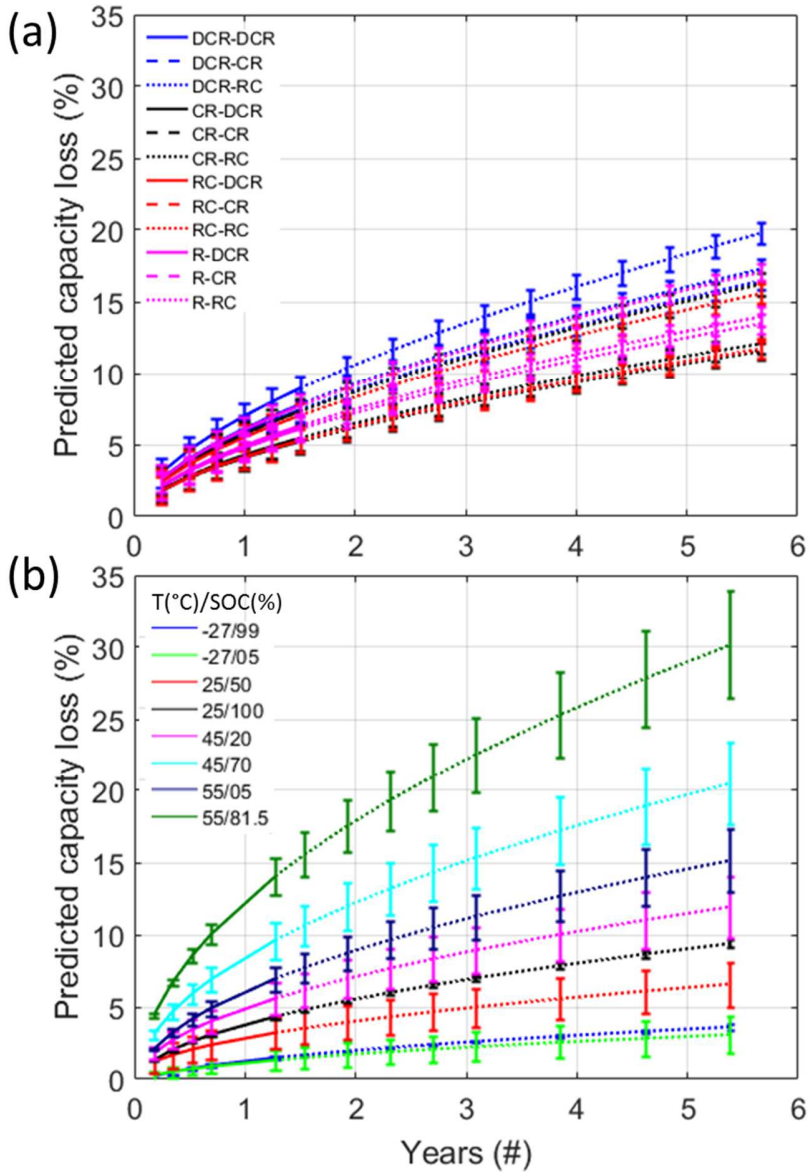



Figure 8: Extrapolation of model parameters for (a) the cycle aging experiment and (b) the calendar aging experiment. Errors bars represent 95% confidence bounds on fitting parameters for both models plus the experimental observed maximum variation between cells.

It has to be noted that these capacity loss predictions assume that the degradation of the battery remains consistent and that neither a second stage of degradation where the

capacity loss is accelerated nor parasitic reactions will occur. Actual battery degradation analysis cannot be reduced to capacity extrapolation alone. Indeed, the resistance increase can start to play a role in capacity retention and also so do silent degradation modes. These silent modes can occur during the initial aging, hidden by more influential mechanisms [25, 58]. If one of these modes becomes predominant, the capacity fading could hastens. These issues need to be considered since a Frisco at *al.* study of similar cells revealed a drastic, unpredictable increase in capacity loss and resistance after 350 cycles at C/3 [44]. The quantification of the degradation modes cannot be done solely from the observation of capacity and resistance changes. One way to quantify the degradation is to analyze the changes in the cell electrochemical behavior using techniques such as incremental capacity or differential voltage [58]. This will be done in our next effort where the cycle and calendar aging incremental capacity signatures will be analyzed and compared in detail to see if the prognosis models to extrapolate capacity loss for longer periods of time, (Figure 8) were indeed reasonable.

## 6 CONCLUSION

In this work, the effect of V2G and G2V strategies were investigated. In addition, the effects of calendar aging on capacity loss, resistance increase, and rate capability were identified.

It was concluded that a V2G step twice a day increased the capacity loss by 75% and the resistance by 10%. This step once a day was found to accelerate the capacity loss by 33% and the resistance increase by 5%. There was, therefore, a clear detrimental impact of performing V2G under these mild conditions (P/4 for an hour) on these cells. Forecasts

based on the measurement results found that V2G implementation would decrease the lifetime of the battery packs to under 5 years.

In contrast, delaying the charge (G2V) over immediate charging had no significant effect (<1%) on capacity retention, and a limited impact on the resistance (<5%). However, this experiment was performed at room temperature where the SOC exercised very little influence on calendar aging. The SOC was found to have a much higher impact at higher temperature. For that reason the results might not apply in warmer climates where delayed G2V should induce less degradation and thus be beneficial.

Calendar aging influenced the cells little enough that it was beneficial to charge the cells twice a day instead of once. Charging twice per day resulted in 5% less capacity loss and similar resistance increase compared to charging just once daily.

In the calendar aging experiments, the capacity loss as a function of time, SOC and temperature was modeled with a double-quadratic equation in which each parameter was weighed by the square root of time. Storage temperature was found to have the biggest impact on capacity loss and rate capability, but the SOC had more influence on the resistance increase. For an EV application, our study shows that EVs using similar cells should not be left fully charged in warmer climates.

Overall, it was found that V2G strategies had a significant deleterious impact on the cell performance, possibly diminishing the lifetime of battery packs to less than 5 years. In contrast, delayed G2V strategies were found to have a negligible effect on cell performance at room temperature, though these strategies could be advantageous in warmer climates. The next step of the investigation is to analyze more deeply the changes in electrochemical behavior presented here using incremental capacity measurements.

## 7 ACKNOWLEDGMENTS

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## 9 FIGURE CAPTIONS

Figure 1: (a) Test matrix, (b) @Work and @Home duration detail, and (c) detail on the 12 duty schedules used in this study on pristine cells.

Figure 2: Calendar aging test matrix for (a) a central composite design and (b) the optimized design.

Figure 3: (a) Average capacity fade and (b) resistance changes vs. equivalent months of cycling; (c) average capacity fade vs. capacity throughput for the 12 duty cycles tested in this work. Error bars indicate the spread of values between the three tested cells per condition.

Figure 4: (a) Average capacity fade, (b) rate capability variations, and (c) resistance variations vs. temperature and SOC for the eight calendar aging conditions tested in this work. Error bars indicate the spread of values between the two tested cells per condition.

Figure 5: (a) Evolution of coded factors with aging, (b) bar plot showing the pace of capacity loss, (c) comparison of the impact on capacity loss of V2G twice a day, V2G once a day, no V2G, regular charging, delayed charging (G2V), 2 charges per day and 1 charge per day, and (d) bar plot of the resistance increase after 18 months of aging.

Figure 6: (a) Capacity loss vs. temperature and SOC after 66 weeks of aging. Red points represents the experimental data and the grey mesh a linear interpolation between them. (b) Evolution of the capacity loss as a function of temperature, SOC and time according to a double-quadratic model. (c) Impact of temperature, SOC and their

combined effect on capacity loss after 66 weeks. (d) Evolution of the six parameters of the quadratic models with aging.

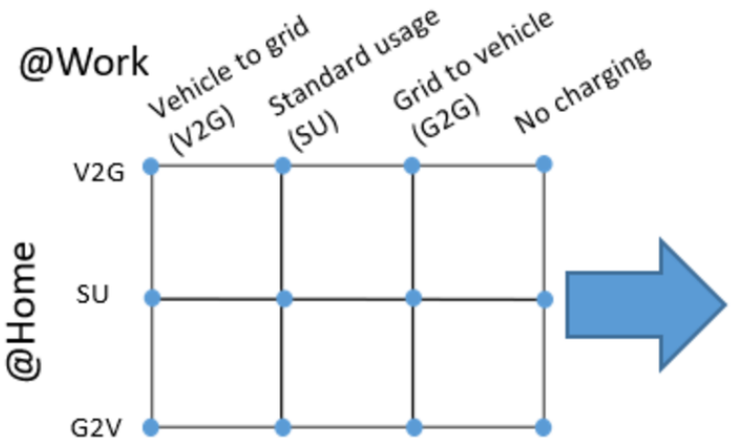
Figure 7: Evolution after 66 weeks of (a) the resistance as a function of SOC and (b) the rate capability as a function of temperature; (c) correlation between capacity loss, rate capability and resistance and (d) incremental capacity curves at C/35 and C/3 for the cell aged at 55°C and 81.5% SOC.

Figure 8: Extrapolation of model parameters for (a) the cycle aging experiment and (b) the calendar aging experiment. Errors bars represent 95% confidence bounds on fitting parameters for both models plus the experimental observed maximum variation between cells.

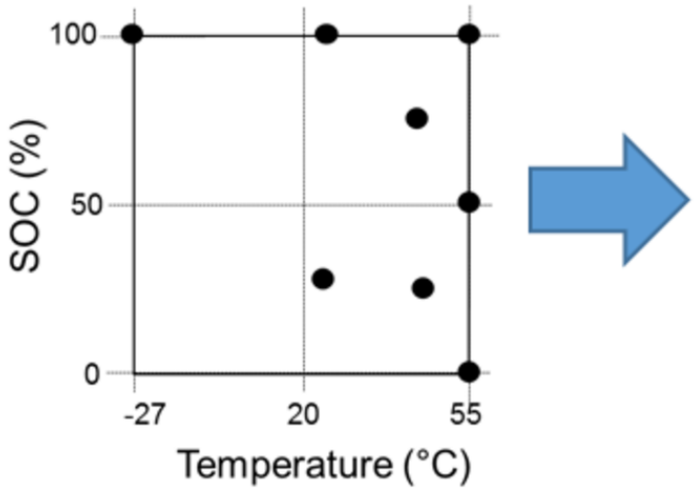
## Experimental matrices

Design of experiment methodology

### Cycle aging experiment

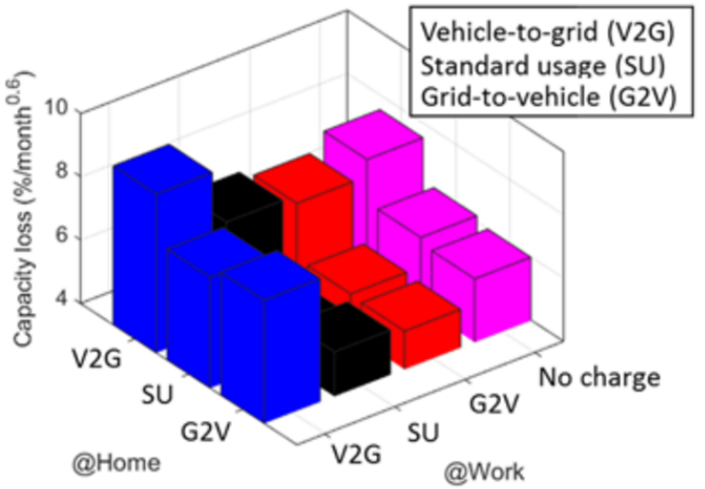


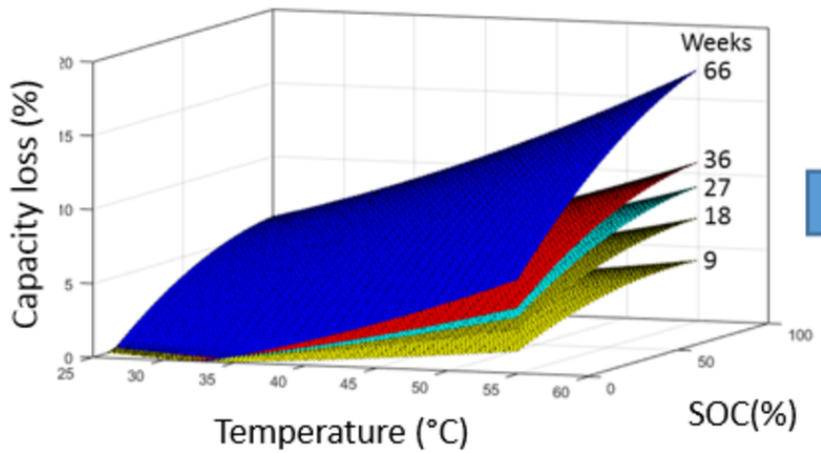

### Calendar aging experiment




**Results:**

V2G 2x1h/day @ P/4 : +75% capacity loss
V2G 1h/day @ P/4 : +33% capacity loss
Charging 2x/day vs. 1x/ day: -5% capacity loss
SU vs. G2V : no significant effect




**Results:**

Significant effect of time, temperature and SOC.
Temperature effect > SOC effect at high values

**Prognosis:**

V2G strategy drastically reduces durability
G2V might be beneficial in warm climates

